

Machine Learning Fundamentals - Exam 2024

MLDM & DSC 1st year Master

Allowed documents: 1 sheet, A4, recto-verso, manuscript, original.
Not programmable calculator allowed.

Exercise 1: Decision Tree

You want to guess whether your friends will join to play beach volley today given the past experiences. You think that 4 elements are important to decide that: the Outlook (Otl), the Temperature (Temp), the Humidity (Hu), and the Wind (Wd).

You have $Otl \in \{\text{Sun (S), Overcast (O), Rain (R)}\}$, $Temp \in \{\text{Warm (W), Mild (M), Cold (C)}\}$, $Hu \in \{\text{High (H), Normal (N), Low (L)}\}$, and $Wd \in \{\text{Strong (S), None (W)}\}$.

Here are a table summarizing the previous days:

Otl	Temp	Hu	Wd	Play?
S	W	H	N	No
S	W	H	S	No
O	W	H	N	Yes
R	M	H	N	Yes
R	C	L	N	Yes
R	C	L	S	No
O	C	L	S	Yes
S	M	H	N	No
S	C	L	N	Yes
R	M	H	S	No

Question 1.1

What are the attributes of this problem? How many data are there?

Question 1.2

You decide to build a decision tree using these data.

- Show graphically and the details of the computation that allows the choice of the root of the tree using the Gini index.
- By using the Gini index, give the graphical output of the algorithm if we fix the maximal depth at 2.

Question 1.3

What is the value of the empirical risk of your learned tree?

Explain in 3 sentences max why having a low empirical risk is always possible with decision trees and why it is not always a relevant metric for Machine Learning.

Question 1.4

Today is a Sunny day, the temperature is 20°C, the humidity is 5%, and there is almost no wind. According to your model, will you play beach volley? Explain the different decisions you did to give your answer.

Exercise 2: Perceptron

Question 2.1

Recall what a perceptron is in max 2 sentences. What is the goal of a learning algorithm for this model?

Question 2.2

We have these data: $x^1=(-1, -1)$, $y^1=-1$; $x^2=(-1, 1)$, $y^2=+1$; $x^3=(2, 2)$, $y^3=+1$

Run the **initial version** of the perceptron learning algorithm on these data.

To present your computation you should use a table like this one:

	init	First loop			Second loop			...
		x^1	x^2	x^3	x^1	x^2	x^3	
w								
b								

And then detail the computation for each cell below the table.

Question 2.3

Draw the data and the obtained linear separator on a plot.

Question 2.4

Without doing any computation, explain in two sentences max (and an eventual plot) what would happen if we add the data $x^4=(-1, 2)$, $y^4=-1$ and run the learning algorithm again.

Exercise 3: HMM 1

A professor has 3 different dices: one D4 that provides a score between 1 and 4, one D6 – the usual dice that gives a score between 1 and 6, and a D10 that turns a score between 1 and 10.

The professor picks at random a dice, throws it, store the score obtain, and repeat the process a random number of times.

In this context, a sequence of observation can be 1, 6, 2, 9, 4.

The objective of this exercise is to model by a HMM the underlying model of the professor's game.

Question 3.1

Represent by a figure the hidden Markov model corresponding to this problem, where the scores are the observable variables, while the dices used are the hidden ones.

Exercise 4: HMM

Suppose that we have a three states (numbered 1 to 3) HMM λ where the observations can be either a or b such that :

$$A = \begin{pmatrix} 0.45 & 0.35 & 0.20 \\ 0.10 & 0.50 & 0.40 \\ 0.15 & 0.25 & 0.60 \end{pmatrix} \quad B = \begin{pmatrix} 1.0 & 0.0 \\ 0.5 & 0.5 \\ 0.0 & 1.0 \end{pmatrix} \quad \pi = \begin{pmatrix} 0.5 \\ 0.3 \\ 0.2 \end{pmatrix}$$

This means that, for instance, $a_{11}=0.45$ and $b_1(a)=1.0$

Question 4.1

Check that $P(ba|\lambda)=0.1075$. You should present the computation in the form of a table, as presented in the course.

Question 4.2

What is the most likely sequence of hidden states for the observation ba ? Again, your computation should be presented in the form of a table.

Question 4.3

Give the obtained HMM when one iteration of the Baum-Welch algorithm is run from this HMM with $\{ba\}$ as a learning sample.